

ABSTRACT:

The main objective of the project is to identify the relationship between social media engagement metrics and product or service sales. The study involves collecting real or sample datasets from various social media platforms and company sales records, followed by cleaning, processing, and analyzing the data using python.

❖ Python libraries such as Pandas and NumPy are used for data manipulation, Matplotlib and seaborn for data visualization, and Scikit-learn for implementing machine learning models like linear regression and decision trees to predict sales based on social media activity. Correlation analysis and time series analysis are also applied to uncover patterns and trends.

❖ By visualizing and interpreting these insights, the project aims to help businesses understand which social media strategies are most effective in driving sales. The outcome of this analysis can support data-driven decision-making, improve marketing ROI, and guide future digital marketing campaigns.

❖ This project not only demonstrates the practical application of data analytics in a real-world scenario but also highlights the growing importance of social media in shaping modern business strategies

